
Lip-Sync Model Selection: Technical Trade-Off Study

Project: AI Lip-Sync Web Service

Deployment Platform: Modal.com

Target Architecture: FastAPI + Decoupled Async Inference

1. Executive Summary

This document provides a comprehensive trade-off analysis between the industry-standard **Wav2Lip** and newer state-of-the-art (SOTA) models like **LivePortrait** and Meta's **MoCha**. Based on the **THEval Framework** which demonstrates an **87%** correlation with human perception and operational constraints on the Modal platform, we have selected **Wav2Lip** for the initial deployment. This choice prioritizes synchronization accuracy, inference speed, and cost-efficiency over high-resolution aesthetic expressiveness.

2. Problem Statement

Input: A source video (MP4) and target audio (WAV/MP3).

Input Constraint Assumptions

To ensure a successful inference pass within the **600-second** Modal timeout and 16GB–24GB VRAM limits, the following constraints are assumed:

1. Video & Facial Geometry

- **Single Subject:** The pipeline assumes a **single, clearly visible face**. Multi-speaker scenarios (where two faces are present) may lead to ambiguous mapping or detection failure.
- **Frontal Orientation:** The subject should maintain a **frontal or semi-frontal pose**.

2. Temporal & Resolution Limits

- **Resolution Ceiling:** Standard Wav2Lip is optimized for **720p (1280x720)**. While 1080p is supported, it is assumed the input will be downscaled or processed with a `--resize_factor` to prevent Out-of-Memory (OOM) errors on budget GPUs like the NVIDIA T4.
- **Clip Duration:** To guarantee completion within the **10-minute Modal timeout**, input audio/video should be limited to **<60 seconds**. (Wav2Lip typically processes at 1.5x–2x real-time on an A10G).
- **Frame Rate Consistency:** Input video is assumed to be a standard **25 or 30 FPS**.

3. Audio Specifications

- **Clean Signal:** The audio must have a high signal-to-noise ratio. Excessive background noise or music can confuse the mel-spectrogram feature extraction.
- **Sampling Rate:** The pipeline assumes/re-samples the target audio to **16kHz mono**, matching the requirements of the pre-trained SyncNet encoder.

4. Environment & Licensing

- **Non-Commercial Intent:** As this uses the LRS2-trained model, all inputs are assumed to be for **educational, research, or demo purposes** (per the Rudrabha/Wav2Lip license).
- **Compute Tier:** It is assumed the service will run on **T4 (16GB VRAM)**, as this provides the best price-to-performance ratio for the GAN-based backbone.

Goal: Generate a new video where the subject's lip movements are perfectly synchronized to the target audio while maintaining the original subject's identity and background stability.

The Challenge: Balancing **lip-sync precision** (SyncNet accuracy) with **visual fidelity** (preventing blurring or artifacts in the mouth region).

3. Evaluation Frameworks & Metrics

To move beyond subjective "it looks good" assessments, this project adopts the benchmarks used in recent 2025 research.

A. The THEval Metric Suite

Currently, simple pixel-matching fails to capture realism. **THEval** breaks evaluation into three core dimensions:

1. **Quality:** Global Aesthetics, Face Quality, Mouth Quality.
2. **Naturalness:** Lip Dynamics, Head Motion, Eyebrow Dynamics.
3. **Synchronization:** Silent Lip Stability, Lip-Sync.

B. Standard Sync Metrics (SyncNet)

We utilize the **Lip-Sync Error Distance (LSE-D)** and **Lip-Sync Error Confidence (LSE-C)** as defined in the original Wav2Lip research:

$$LSE-D = \frac{1}{N} \sum_{i=1}^N \|f_{\text{video}}(i) - f_{\text{audio}}(i)\|$$

- **LSE-D:** Lower is better (measures embedding distance).

- **LSE-C:** Higher is better (measures the "expert" discriminator's confidence in the match).

C. MoCha-Bench (Meta 2025)

Meta’s **MoCha-Bench** adds a "cinematic" layer to evaluation, focusing on whether the character’s expressions align with the **emotional tone** of the audio, not just the phonemes.

4. Deep-Dive Model Comparison

Criteria	Wav2Lip	LivePortrait	Analysis
GitHub Stats	12.7k stars	17.5k stars	LivePortrait has a stronger, active community.
Last Update	Aug 2020 (Legacy)	June 2025 (Active)	Wav2Lip is legacy/unmaintained.
Python Version	3.6+	3.10+	LivePortrait requires a modern stack.
Dependencies	PyTorch, FFmpeg	PyTorch 2.3+, Transformers	LivePortrait is significantly more complex.
Input Constraints	720p, Various Ratios	512x512, 1:1 ratio	Wav2Lip is more flexible with source video.
GPU Support	CUDA (Optimized for T4)	CUDA + Apple Silicon	LivePortrait requires higher VRAM (A10G+).
Commercial Use	Non-commercial	Not explicitly restricted	Critical licensing difference (LRS2 dataset).

Technical Decision Matrix

1. Quality & Synchronization (30% weight)

- **Wav2Lip:**
 - Lower visual resolution (limited to 720p).
 - Known for "blurring" artifacts in the mouth region.
 - **Score: 6/10**
- **LivePortrait:**
 - Superior global aesthetics and identity preservation.
 - General portrait animation; not exclusively optimized for phoneme-to-viseme sync.
 - **Score: 7/10**

Winner: LivePortrait (Higher visual fidelity but lower sync-focus).

2. Speed & Performance (25% weight)

- **Wav2Lip:**
 - GAN-based architecture allows for **~1-3 sec/frame** on a T4.
 - Highly predictable latency.
 - **Score: 7/10**
- **LivePortrait:**
 - Heavy diffusion/transformer dependencies increase cold-start and inference time.
 - Video-to-video processing is computationally expensive.
 - **Score: 5/10**

Winner: Wav2Lip (Essential for meeting the \$30 Modal budget and low-latency requirements).

3. GPU Requirements (20% weight)

- **Wav2Lip:**
 - **VRAM Status:** No explicit VRAM requirement stated in official docs.
 - **Legacy Advantage:** Older GAN-based architecture leads to significantly lower hardware requirements.
 - **Estimated: 4-6 GB VRAM.**
 - **Score: 7/10**
- **LivePortrait:**
 - **VRAM Status:** No explicit VRAM stated, but architectural complexity is higher.
 - **Resource Heavy:** Dependencies like PyTorch 2.3 and modern transformers necessitate higher memory overhead.
 - **Estimated: 8-12 GB VRAM.**
 - **Score: 5/10**

Winner: Wav2Lip (14 vs 10 weighted points)

4. Ease of Deployment (15% weight)

- **Wav2Lip:**
 - **Simplicity:** Minimal dependencies (primarily PyTorch and FFmpeg).
 - **Portability:** Pre-trained models are easily accessible via Google Drive downloads.
 - **Legacy Debt:** 5 years of age suggests potential compatibility issues with Python 3.11+ environments.
 - **Stagnant:** No active maintenance or official Docker support.
 - **Score: 6/10**
- **LivePortrait:**
 - **Modern Ecosystem:** Built on a 2025-ready stack with HuggingFace integration.
 - **Accessibility:** Provides a Gradio UI for immediate testing.
 - **Complexity:** Requires specialized setup for certain features (e.g., X-Pose for animal models).
 - **Pre-processing:** Demands specific input constraints (512x512) for optimal results.
 - **Score: 6/10**

Winner: Tie (9 vs 9 weighted points)

5. Documentation & Support (10% weight)

- **Wav2Lip:**
 - **Clarity:** The README remains a clear, functional guide with reproducible examples.
 - **Support Gap:** No updates or issue resolutions since August 2020.
 - **Commercial Pivot:** The README primarily redirects users to a commercial service (**Sync.so**).
 - **Score: 5/10**
- **LivePortrait:**
 - **High Engagement:** Active development (latest updates in Jan/June 2025) with over **1.8k forks**.
 - **Tooling:** Native Gradio UI and widespread community wrappers (ComfyUI nodes).
 - **Industry Adoption:** Already utilized by major platforms like Kuaishou and Douyin.
 - **Score: 8/10**

Winner: LivePortrait (8 vs 5 weighted points)

Final Weighted Score

Model	Quality (30%)	Speed (25%)	GPU (20%)	Deployment (15%)	Docs (10%)	TOTAL
Wav2Lip	18	17.5	14	9	5	63.5/100
LivePortrait	21	12.5	10	9	8	60.5/100

Final Recommendation: Wav2Lip

Reasoning:

- Scope Fit:** Faster inference translates to lower Modal costs.
- Resource Efficiency:** Runs optimally on an **NVIDIA T4 (\$0.59/hr)** vs. requiring an expensive A100.
- Synchronization Gold Standard:** Despite lower resolution, it remains the benchmark for academic lip-sync accuracy.

5. Formal References

Primary Research Papers

- Prajwal, K. R., Mukhopadhyay, R., Namboodiri, V. P., & Jawahar, C. V. (2020).** *A Lip Sync Expert Is All You Need for Speech to Lip Generation In The Wild*. Proceedings of the 28th ACM International Conference on Multimedia (ACM MM), 484–492. <https://doi.org/10.1145/3394171.3413532>
- Quignon, N., Chopin, B., Wang, Y., & Dantcheva, A. (2025).** *THEval: Evaluation Framework for Talking Head Video Generation*. arXiv preprint arXiv:2511.04520. (Introduces the 8-metric benchmark for talking heads).
- Guo, J., Zhang, D., et al. (2024).** *LivePortrait: Efficient Portrait Animation with Stitching and Retargeting Control*. arXiv:2407.03168.
- Wei, C., et al. (2025).** *MoCha: Towards Movie-Grade Talking Character Synthesis*. arXiv:2503.23307. (Defines the current SOTA for cinematic character generation).
- Yaman, D., et al. (2024b).** *Audio-driven Talking Face Generation with Stabilized Synchronization Loss*. European Conference on Computer Vision (ECCV). (Discusses instabilities in traditional SyncNet metrics).

Technical Manuals & Industry Sources

- Modal Documentation (2025).** *Serverless GPU Infrastructure & Cold-Start Optimization*. modal.com/docs

- **Hugging Face Datasets.** *MoChaBench: A benchmark tailored for Talking Characters generation tasks.* <https://huggingface.co/datasets/CongWei1230/MoChaBench>